

SPEC-COLLAPSE ATLAS

Does pooling a multiverse manufacture false robustness?

QUESTION

When a multiverse meta-analysis is summarised by pooling its specifications, does that summary overstate how robust the underlying conclusion really is?

METHOD

473 Cochrane meta-analyses

Pairwise70 corpus · binary + continuous · $k \geq 3$

re-analysed across 36 specifications

estimator × interval × outlier × publication-bias



Naive IV-RE pool

specs treated as independent

Calibrated weighted-likelihood interval

HEADLINE

“robust” conclusions

88%



33%

naive

calibrated

FALSE-ROBUSTNESS RATE

55%

260 of 473 reviews flip robust → fragile

under a simulated null effect

naive CIs $\approx$ 1/8 calibrated width (median 0.12×)

72-81%

naive false-positive rate (nominal 5%)

SCOPE

Estimand = false-robustness rate (binary significance agreement). This analysis covers significance reversal, not effect-magnitude shifts. IV-RE pooling across specifications treats re-analyses of one dataset as independent evidence, manufacturing precision the data do not support.